Fact-Checking in NLP

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Lesson Plan

Introduction

Claim Detection

Claim Verification
- Evidence Retrieval & Selection
- Veracity Prediction

Conclusion

Goal of this lecture

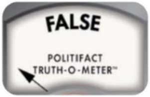
- Provide an overview of tasks related to fact-checking in NLP
- Discuss interesting & common approaches
- Consider ongoing challenges

Lesson Plan

Real-world example

Donald Trump

stated on September 29, 2020 in the first presidential debate:



In manufacturing, "I brought back 700,000 jobs. (Obama and Biden) brought back nothing."

IF YOUR TIME IS SHORT

- Trump is wrong about the job gains on his watch; the actual increase is about 450,000 prior to the pandemic.
- As for Obama and Biden, they saw gains of 916,000 if you start counting with the recovery from the Great Recession, which is the fairest comparison if you also ignore the losses under Trump during the pandemic.

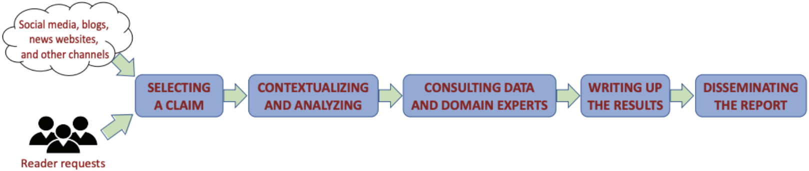
Image from [6]

Related tasks

- Rumor detection
- Misinformation detection
- Fake-news detection
- Stance detection

How do humans do fact-checking?

The process:



From [13]

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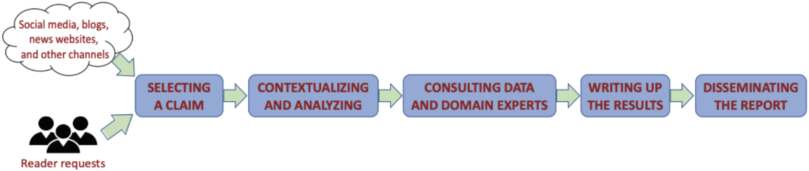
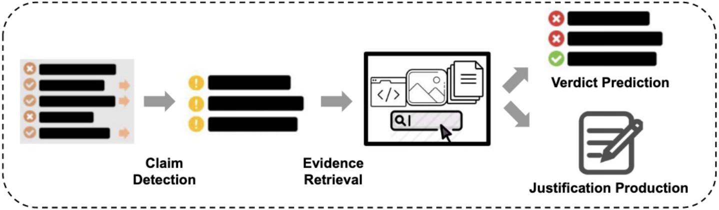


Potential challenges:

Research Questions	Key Themes
RQ1: What practices are involved in the work of professional fact-checkers?	Monitoring multiple channels (**21**) Dealing with reader requests (**21**) Filtering and prioritizing claims (**21**) Selecting claims to verify (**21**) Contextualizing and analyzing claims (**21**) Consulting data and domain experts (**17**) Reviewing write-ups (**18**)

From [13]

NLP Fact-checking pipeline



- Typically 3 steps
  - Claim detection
  - Evidence retrieval
  - Verdict prediction

From [6]

Lesson Plan

What is a claim?

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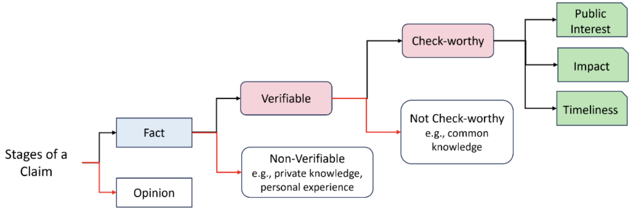

Figure 3: Stages of a claim

Black arrows in the figure indicate claims and red arrows indicate non-claims

- Factual
- Checkworthy
- Verifiable

From [14]

Detailed Claim breakdown

Category	Subcategory	Counts	Example
Not a claim		54.8%	"Give it all to them, I really don't mind."
Other	Other other	10.4%*	"Molly gives so much of who she is away throughout the film."
	Support/policy	5.5%*	"He has advocated for a junk food tax."
	Quote	4.7%*	"The Brexit secretary said he would guarantee free movement of bankers."
	Trivial claim	1.6%*	"It was a close call."
	Voting record	0.7%*	"She just lost a parliamentary vote."
	Public opinion	0.4%*	"A poll showed that most people who voted Brexit were concerned with immigration."
	Definition	0.0%*	"The unemployed are only those actively looking for work."
Quantity	Current value	9.9%	"1 in 4 people wait longer than 6 weeks to see a doctor."
	Changing quantity		
	Comparison		
	Ranking		
Prediction	Hypothetical statements	4.4%	"The IFS says that school funding will have fallen by 5% by 2019."
	Claims about the future		
Personal experience	Uncheckable	3.0%	"I can't save for a deposit"
Correlation/ causation	Correlation	2.6%	"Tetanus vaccine causes infertility"
	Causation		
	Absence of a link		
Laws/rules of operation	Public institution-al procedures	1.9%	"The UK allows a single adult to care for fewer children than other European countries."
	Rules/rule changes		

Annotation scheme from [7]

Claim detection – the task

- Classification or ranking
  - Given: sentence
  - Output: is it a claim?
    - is it *checkworthy*?

      Not possible to check everything

An example: SciClops [17] (Baselines)

- Grammar-based:
  $(root(s) \in RV) \land ((nsubj(s) \in E) \lor (dobj(s) \in E)) \implies (s \in Claims)$
  - *RV:* reporting verbs (ex: "say", "claim")
  - *E:* entities (ex: people, organizations)

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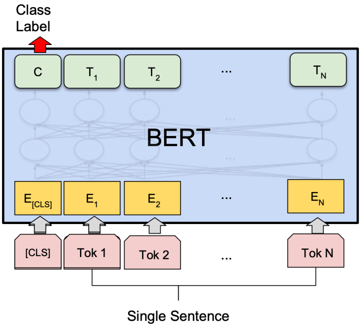
- Context-based:
  $$(\exists p : sim(s, p) \cdot pop(p) \geq threshold) \implies (s \in Claims)$$
  - *sim:* embedding-based cosine similarity
  - *pop:* popularity, based on re-posting/likes

An example: SciClops [17] (Method)



BERT-based classifiers

- Fine-tuned binary classifier

Image from: https://wandb.ai/mukilan/BERT_Sentiment_Analysis/reports/An-Introduction-to-BERT-And-How-To-Use-It--VmlldzoyNTIyOTA1

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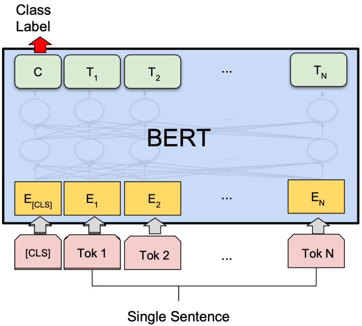

BERT-based classifiers

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Variations: pre-train LM on domain-specific corpora

- Scientific papers, news headlines, or both

Image from: https://wandb.ai/mukilan/BERT_Sentiment_Analysis/reports/An-Introduction-to-BERT-And-How-To-Use-It--VmlldzoyNTIyOTA1

An example: SciClops [17] (Results)

Automatic evaluation

		Generic Dataset *ACC*	Scientific Dataset *ACC*
Baseline	*Grammar-Based*	50.4%	52.3%
	Context-Based	49.5%	50.2%
	Random Forest	74.7%	75.6%
	BERT	**82.2%**	81.0%
SciClops	*SciBERT*	81.5%	80.6%
	NewsBERT	82.0%	80.0%
	SciNewsBERT	81.1%	**81.2%**

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Human evaluation

		Weak Agreement P	Weak Agreement R	Weak Agreement F1	Strong Agreement P	Strong Agreement R	Strong Agreement F1
Baseline	*Grammar-Based*	51.8%	70.4%	59.9%	40.4%	28.0%	33.1%
	Context-Based	44.6%	49.6%	47.0%	24.5%	45.1%	31.8%
	Random Forest-gen	52.1%	70.4%	59.9%	43.7%	**80.5%**	56.7%
	Random Forest-sci	56.7%	54.4%	55.5%	43.3%	44.8%	44.1%
	BERT-gen	50.8%	50.4%	50.6%	33.5%	68.3%	45.0%
	BERT-sci	78.7%	38.4%	51.6%	79.2%	51.2%	62.2%
SciClops	*NewsBERT-gen*	55.0%	48.8%	51.7%	38.9%	62.2%	47.9%
	NewsBERT-sci	76.9%	40.0%	52.6%	74.2%	56.1%	63.9%
	SciBERT-gen	48.8%	66.4%	56.3%	32.8%	72.0%	45.0%
	SciBERT-sci	48.8%	66.4%	56.2%	**86.5%**	39.1%	53.8%
	SciNewsBERT-gen	49.8%	**80.0%**	**61.3%**	38.8%	78.0%	51.8%
	SciNewsBERT-sci	**84.4%**	30.4%	44.7%	82.7%	52.4%	**64.2%**

- Human evaluation gives more nuanced view of the results

Observations & ongoing challenges

- Checkworthiness criteria - domain-specific
  - Science: causal relation? prevalence?
  - Green claims: implicit vs. explicit

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- Limited domain & language coverage [6]

Dataset	Type	Input	#Inputs	Evidence	Verdict	Sources	Lang
CredBank (Mitra and Gilbert, 2015)	Worthy	Aggregate	1,049	Meta	5 Classes	Twitter	En
Weibo (Ma et al., 2016)	Worthy	Aggregate	5,656	Meta	2 Classes	Twitter/Weibo	En/Ch
PHEME (Zubiaga et al., 2016)	Worthy	Individual	330	Text/Meta	3 Classes	Twitter	En/De
RumourEval19 (Gorrell et al., 2019)	Worthy	Individual	446	Text/Meta	3 Classes	Twitter/Reddit	En
DAST (Lillie et al., 2019)	Worthy	Individual	220	Text/Meta	3 Classes	Reddit	Da
Suspicious (Volkova et al., 2017)	Worthy	Individual	131,584	✗	2/5 Classes	Twitter	En
CheckThat20-T1 (Barrón-Cedeño et al., 2020)	Worthy	Individual	8,812	✗	Ranking	Twitter	En/Ar
CheckThat21-T1A (Nakov et al., 2021b)	Worthy	Individual	17,282	✗	2 Classes	Twitter	Many
Debate (Hassan et al., 2015)	Worthy	Statement	1,571	✗	3 Classes	Transcript	En
ClaimRank (Gencheva et al., 2017)	Worthy	Statement	5,415	✗	Ranking	Transcript	En
CheckThat18-T1 (Atanasova et al., 2018)	Worthy	Statement	16,200	✗	Ranking	Transcript	En/Ar
CitationReason (Redi et al., 2019)	Checkable	Statement	4,000	Meta	13 Classes	Wikipedia	En
PolitiTV (Konstantinovskiy et al., 2021)	Checkable	Statement	6,304	✗	7 Classes	Transcript	En

Observations & ongoing challenges

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- Context

Challenge example: de-contextualization

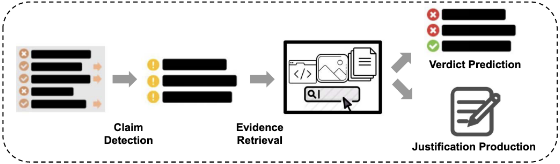
- Without context, claims lack meaning

Coreference Resolution
Sentence: He has pubicly stated that he sympathized with their cause and even hinted that he would provide them with American resources should they be in need during his 2008 State of the Union address. **Decontextualised sentence**: President Obama has pubicly stated that he sympathized with their cause and even hinted that he would provide them with American resources should they be in need during his 2008 State of the Union address.
Global Scoping
Sentence: During the attack, Capitol Police made the request again. **Decontextualised sentence**: During the attack on Washington D.C., Capitol Police made the request again.
Bridge Anaphora
Sentence: The government does not have proper storage facilities for stocking such a large amount of excess grain. **Decontextualised sentence**: The government of India does not have proper storage facilities for stocking such a large amount of excess grain.

From [5]

Lesson Plan

Overview



- Most of the work on fact-checking is here
- Two components, usually together
  - Evidence retrieval & selection
  - Veracity prediction
  - Sometimes also: justification production

From [6]

Lesson Plan

Evidence Retrieval: the task

- Given: a claim
- Find: relevant evidence
  - Documents (e.g., news articles, scientific abstracts)
  - Sentences

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- How?
    - Score & rank possible evidence then take top k
    - Information Retrieval (IR) is a whole area of computer science

Sources of evidence

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From [2]

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  - From a homogenous source
    (Wikipedia, new articles, summaries, scientific abstracts, etc.)
  - Unstructured textual data
    - Sometimes also semi-structured data – tables
      (Wikipedia tables, scientific article tables)

Claim: In the 2018 Naples general election, Roberto Fico, an Italian politician and member of the Five Star Movement, received 57,119 votes with 57.6 percent of the total votes.

Evidence:

Page: `wiki/Roberto_Fico`

$\mathbf{e}_1$(Electoral history):

2018 general election: Naples -Fuorigrotta		
Candidate	Party	Votes
Roberto Fico	Five Star	61,819
Marta Schifone	Centre-right	21,651
Daniela Iaconis	Centre-left	15,779

Verdict: Refuted

From [2]

Evidence Retrieval: basic approach

- Document retrieval: similarity scoring function $\rightarrow$ ranking [9]
  - TF-IDF

    Uses: Term Frequency (TF), Inverse Document Frequency (IDF)
  - BM25

    Uses: TF, IDF, Document Length

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- Tables?
  - Often given (e.g., as in [12])
  - Prompting to choose possible (SPIQA [15])

Evidence selection (rationale selection)

- **Pairwise sentence classification** (as in [19])
  - BERT-like input:
    `[CLS] sentence [SEP] claim [SEP]`
  - Score:
    probability the sentence is evidence related to the claim
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- **Token-level scoring** & aggregation [18]
  - Accounts for document context
    claim: *St. Anger is the first studio album by Metallica.*
    evidence: *It was released in June 2003 as the lead single from their eighth studio album of the same name.*

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    claim: St. Anger is the first studio album by Metallica.
    evidence: It was released in June 2003 as the lead single from their eighth studio album of the same name.
- Similar to: claim detection, QA answer-span extraction

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Veracity prediction task

- Given: a claim & evidence
- Predict: truth of claim in *relation* to the evidence
  - Usually *supported/refuted/not enough evidence*

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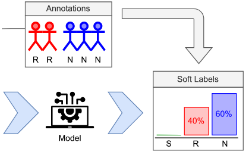
  - Other variations
    - Soft labels



    - Additional labels
      Ex: *conflicting evidence/cherry-picking* [16]

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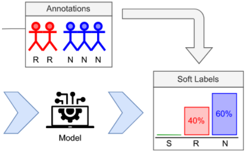

    - Additional labels
      Ex: *conflicting evidence/cherry-picking* [16]
  - Better schema - open area of research

Basic approach to veracity prediction

- Sentence-pair classification
  - Encoder only:
    `[CLS] evidence [SEP] claim [SEP]` $\rightarrow h_{[CLS]}$
  - Encoder-decoder:

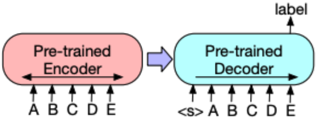

Image from [10]

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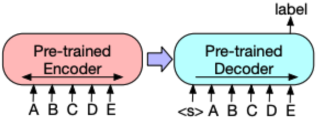


- Tabular data $\rightarrow$ flatten/linearize
  $T^* = \texttt{[HEAD]}, c_1, \cdots, c_N, \texttt{[ROW]}, 1, r_1, \texttt{[ROW]}, 2, r_2, \ldots, r_M$ [11]

Image from [10]

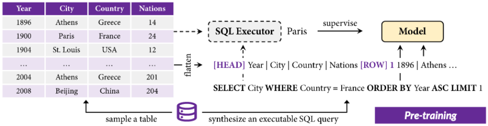
Easy improvements for veracity prediction?

Better or domain-specific LMs

- "Better" LMs
  - BERT & BART → Llama, Alpaca, Vicuna, etc. (as in [12])

- Domain-specific LMs
  - SciBERT (as in [17])
  - Table fine-tuned LMs

    (ex: TAPEX [11])



Domain specific isn't always better

Models		# of Para.	Zero-shot		In-Context	
			2-class	3-class	2-class	3-class
I. Table-based LLMs	TAPAS-large (Tabfact) (Herzig et al., 2020)	340M	50.30	—	—	—
	TAPEX-large (Tabfact) (Liu et al., 2022b)	400M	56.06	—	—	—
	TAPEX-Zero-large (Liu et al., 2023b)	780M	48.28	29.72	42.44	23.47
	TAPEX-Zero-XL (Liu et al., 2023b)	3B	49.77	34.30	42.12	25.62
II. Encoder–Decoder LLMs	Flan-T5-base (Chung et al., 2022)	250M	47.38	26.56	44.82	24.09
	Flan-T5-large (Chung et al., 2022)	780M	51.58	32.55	49.62	27.30
	FLan-T5-XL (Chung et al., 2022)	3B	52.41	**38.05**	48.05	29.21
	Flan-T5-XXL (Chung et al., 2022)	11B	59.60	34.91	**60.48**	34.04
III. Open source LLMs	Alpaca-7B (Taori et al., 2023)	7B	37.22	27.59	40.46	28.95
	Vicuna-7B (Chiang et al., 2023)	7B	**63.62**	32.47	50.35	34.26
	Vicuna-13B (Chiang et al., 2023)	13B	41.82	29.63	55.11	**35.16**
	LLaMA-7B (Touvron et al., 2023)	7B	49.05	32.26	45.24	27.17
	LLaMA-13B (Touvron et al., 2023)	13B	53.97	37.18	44.39	32.66

From [12]

Veracity prediction with prompting

```
Caption: Table 4: Scores for different training objectives on the linguistic probing tasks.

Table:
|| Method | Depth| BShift| SubjNum | Tense | CoordInv | Length | ObjNum | TopConst | SOMO | WC ||
|| CMOW-C | 36.2 | 66.0 | 81.1 | 78.7 | 61.7 | 83.9 | 79.1 | 73.6 | 50.4 | 66.8 ||
|| CMOW-R | 35.1 | 70.8 | 82.0 | 80.2 | 61.8 | 82.8 | 79.7 | 74.2 | 50.7 | 72.9 ||
|| CBOW-C | 34.3 | 50.5 | 79.8 | 79.9 | 53.0 | 75.9 | 79.8 | 72.9 | 48.6 | 89.0 ||
|| CBOW-R | 33.0 | 49.6 | 79.3 | 78.4 | 53.6 | 74.5 | 78.6 | 72.0 | 49.6 | 89.5 ||

Claim: While CMOW-R and CMOW-C perform comparably on most probing tasks,
CMOW-C yields 5 points higher scores on WordContent and BigramShift.

Question: Is the above claim true or false? Let us think step by step.

Reasoning:
To verify this claim, we need to compare the performance of CMOW-R
and CMOW-C on WordContent and BigramShift.
The performance of CMOW-R on WordContent is 72.9 (table[2,10]).
The performance of CMOW-C on WordContent is 66.8 (table[1,10]).
The performance of CMOW-R on BigramShift is 70.8 (table[2,2]).
The performance of CMOW-C on BigramShift is 66.0 (table[1,2]).
Therefore, the performance of CMOW-C is higher than
the performance of CMOW-R on WordContent by 66.8 - 72.9 = -6.1.
The performance of CMOW-C is lower than
the performance of CMOW-R on BigramShift by 66.0 - 70.8 = -4.8.
Therefore, the Claim is false.

Answer:
Therefore, the claim is false.

(··· more in-context examples here ···)
```

From [12]

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|| CMOW-R | 35.1 | 70.8 | 82.0 | 80.2 | 61.8 | 82.8 | 79.7 | 74.2 | 50.7 | 72.9 ||
|| CBOW-C | 34.3 | 50.5 | 79.8 | 79.9 | 53.0 | 75.9 | 79.8 | 72.9 | 48.6 | 89.0 ||
|| CBOW-R | 33.0 | 49.6 | 79.3 | 78.4 | 53.6 | 74.5 | 78.6 | 72.0 | 49.6 | 89.5 ||

Claim: While CMOW-R and CMOW-C perform comparably on most probing tasks,
CMOW-C yields 5 points higher scores on WordContent and BigramShift.

Question: Is the above claim true or false? Let us think step by step.

Reasoning:
To verify this claim, we need to compare the performance of CMOW-R
and CMOW-C on WordContent and BigramShift.
The performance of CMOW-R on WordContent is 72.9 (table[2,10]).
The performance of CMOW-C on WordContent is 66.8 (table[1,10]).
The performance of CMOW-R on BigramShift is 70.8 (table[2,2]).
The performance of CMOW-C on BigramShift is 66.0 (table[1,2]).
Therefore, the performance of CMOW-C is higher than
the performance of CMOW-R on WordContent by 66.8 - 72.9 = -6.1.
The performance of CMOW-C is lower than
the performance of CMOW-R on BigramShift by 66.0 - 70.8 = -4.8.
Therefore, the Claim is false.

Answer:
Therefore, the claim is false.

(··· more in-context examples here ···)
```

- Prompting techniques don't necessarily improve models

IV. Close source LLMs	InstructGPT (Ouyang et al., 2022)	175B	68.44	41.41	68.10	41.58
	InstructGPT+CoT (Ouyang et al., 2022)	175B	—	—	68.46	42.60
	PoT (Chen et al., 2022)	175B	—	—	63.79	—
	GPT-4 (OpenAI, 2023)	—	78.22	64.80	77.98	63.21
	GPT-4+CoT (OpenAI, 2023)	—	—	—	76.85	62.77
	Human	—	—	—	92.40	84.73

From [12]

Neurosymbolic veracity prediction – ProofVer [9]

- Intuition: veracity as textual entailment
  - *supported* ≈ entailed by evidence
  - *refuted* ≈ refuted by evidence

Claim

Little Dorrit is a short story by Charles Dickens

Evidence

Little Dorrit is a novel by Charles J H Dickens, originally published in serial form between 1855 and 1857

- Exploit *lexical* entailment for explainability

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- Approach
  - Chunk & align
  - Identify chunk relations ⇝ proof
  - Proof as instructions for DFA ⇝ prediction

ProofVer [9] – some details

Natural logic [3]

- Comparison
  - Equivalence $\equiv$: *by Charles Dickens* $\equiv$ *by Charles J H Dickens*
  - Negation $\curlywedge$: *by Charles Dickens* $\curlywedge$ *not by Charles Dickens*
  - Alternation $\downharpoonleft\upharpoonright$: *by Charles Dickens* $\downharpoonleft\upharpoonright$ *by Emily Dickinson*

    short story $\downharpoonleft\upharpoonright$ *novel*

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Natural logic [3]

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    short story $\downharpoonleft\upharpoonright$ *novel*
- Entailment
  - Forward $\sqsubseteq$: *by Charles Dickens* $\sqsubseteq$ *by Dickens*
  - Reverse $\sqsupseteq$: *story* $\sqsupseteq$ *short story*

ProofVer[9] – more details

Claim

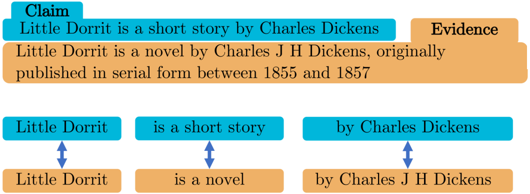
Little Dorrit is a short story by Charles Dickens

Evidence

Little Dorrit is a novel by Charles J H Dickens, originally published in serial form between 1855 and 1857

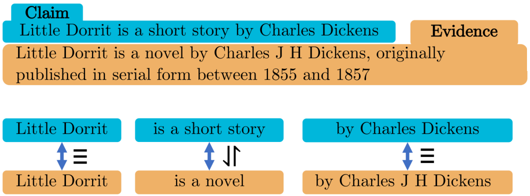
- Approach

ProofVer[9] – more details



- Approach
  - Chunk & align

ProofVer[9] – more details



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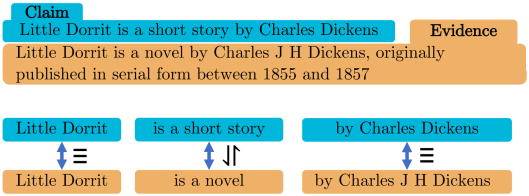
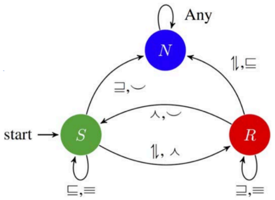
ProofVer[9] – more details





- Approach
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  - Proof as instructions for DFA

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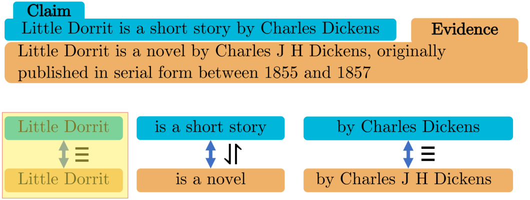

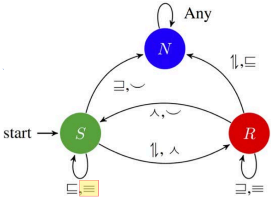

- Approach
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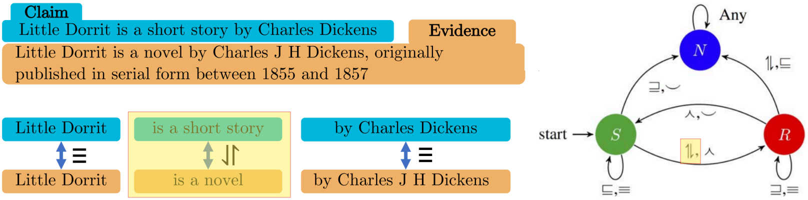
  - Identify chunk relations $\rightsquigarrow$ proof
  - Proof as instructions for DFA

ProofVer[9] – more details



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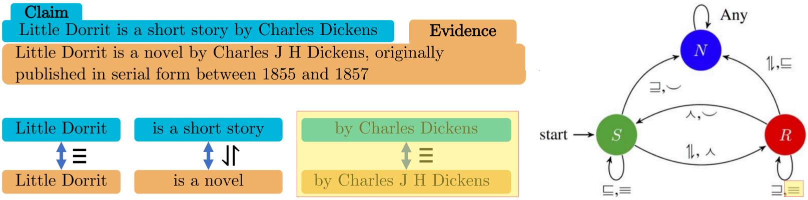

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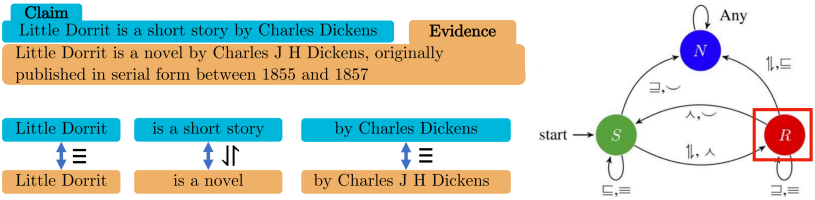

- Approach
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  - Proof as instructions for DFA ⇝ prediction

Other approaches to explainability

Explanation generation as summarisation

- Extractive [4]

Claim: The last major oil spill from a drilling accident in America happened over 40 years ago in 1969.

Ruling Comments: (...) The last major oil spill from a drilling accident in America happened over 40 years ago in 1969.
(...) The largest in volume was the Santa Barbara spill of 1969 referenced by Murdock and Johnson, in which an estimated 100,000 barrels of oil spilled into the Pacific Ocean, according to the API. The Santa Barbara spill was so big it ranked seventh among the 10 largest oil spills caused by marine well blowouts in the world, the report states. Two other U.S. spills, both in 1970, rank eighth and 10th. Fourteen marine blowouts have taken place in the U.S. between 1969 and 2007. Six of them took place after 1990 and spilled a total of nearly 13,700 barrels.
(...) We interviewed three scientists who said that the impact of a spill has little to do with its volume. Scientists have proven that spills far smaller than Santa Barbara's have been devastating.

Justification: While the nation's largest oil well blowout did take place in 1969, it's not factually correct to call it the "last major oil spill". First of all, two of the largest blowouts in the world took place in the U. S. the following year. More importantly, experts agree that spills far smaller in volume to the 1969 disaster have been devastating. From a scientific perspective, Johnson's decision to single out the 1969 blowout as the last "major" one makes no sense.

Image from [4]

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(...) We interviewed three scientists who said that the impact of a spill has little to do with its volume. Scientists have proven that spills far smaller than Santa Barbara's have been devastating.

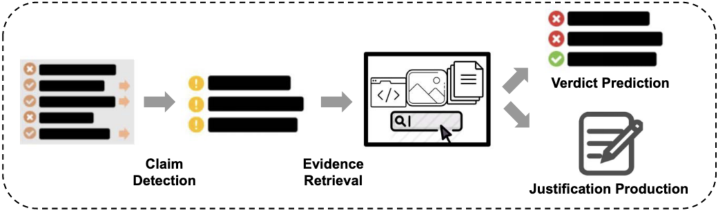
- Abstractive [8]

Justification: While the nation's largest oil well blowout did take place in 1969, it's not factually correct to call it the "last major oil spill". First of all, two of the largest blowouts in the world took place in the U. S. the following year. More importantly, experts agree that spills far smaller in volume to the 1969 disaster have been devastating. From a scientific perspective, Johnson's decision to single out the 1969 blowout as the last "major" one makes no sense.

Image from [4]

Lesson Plan

Summary



- 3 main components to fact-checking
  - Most of the work is concentrated on the latter ones
- Explainability a growing area
- Many remaining challenges

So many datasets

Dataset	Input	#Inputs	Evidence	Verdict	Sources	Lang
CrimeVeri (Bachenko et al., 2008)	Statement	275	✗	2 Classes	Crime	En
Politifact (Vlachos and Riedel, 2014)	Statement	106	Text/Meta	5 Classes	Fact Check	En
StatsProperties (Vlachos and Riedel, 2015)	Statement	7,092	KG	Numeric	Internet	En
Emergent (Ferreira and Vlachos, 2016)	Statement	300	Text	3 Classes	Emergent	En
CreditAssess (Popat et al., 2016)	Statement	5,013	Text	2 Classes	Fact Check/Wiki	En
PunditFact (Rashkin et al., 2017)	Statement	4,361	✗	2/6 Classes	Fact Check	En
Liar (Wang, 2017)	Statement	12,836	Meta	6 Classes	Fact Check	En
Verify (Baly et al., 2018)	Statement	422	Text	2 Classes	Fact Check	Ar/En
CheckThat18-T2 (Barrón-Cedeño et al., 2018)	Statement	150	✗	3 Classes	Transcript	En
Snopes (Hanselowski et al., 2019)	Statement	6,422	Text	3 Classes	Fact Check	En
MultiFC (Augenstein et al., 2019)	Statement	36,534	Text/Meta	2–27 Classes	Fact Check	En
Climate-FEVER (Diggelmann et al., 2020)	Statement	1,535	Text	4 Classes	Climate	En
SciFact (Wadden et al., 2020)	Statement	1,409	Text	3 Classes	Science	En
PUBHEALTH (Kotonya and Toni, 2020b)	Statement	11,832	Text	4 Classes	Fact Check	En
COVID-Fact (Saakyan et al., 2021)	Statement	4,086	Text	2 Classes	Forum	En
X-Fact (Gupta and Srikumar, 2021)	Statement	31,189	Text	7 Classes	Fact Check	Many
cQA (Mihaylova et al., 2018)	Answer	422	Meta	2 Classes	Forum	En
AnswerFact (Zhang et al., 2020)	Answer	60,864	Text	5 Classes	Amazon	En
NELA (Horne et al., 2018)	Article	136,000	✗	2 Classes	News	En
BuzzfeedNews (Potthast et al., 2018)	Article	1,627	Meta	4 Classes	Facebook	En
BuzzFace (Santia and Williams, 2018)	Article	2,263	Meta	4 Classes	Facebook	En
FA-KES (Salem et al., 2019)	Article	804	✗	2 Classes	VDC	En
FakeNewsNet (Shu et al., 2020)	Article	23,196	Meta	2 Classes	Fact Check	En
FakeCovid (Shahi and Nandini, 2020)	Article	5,182	✗	2 Classes	Fact Check	Many

Dataset	Input	#Inputs	Evidence	Verdict	Sources	Lang
KLinker (Ciampaglia et al., 2015)	Triple	10,000	KG	2 Classes	Google/Wiki	En
PredPath (Shi and Weninger, 2016)	Triple	3,559	KG	2 Classes	Google/Wiki	En
KStream (Shiralkar et al., 2017)	Triple	18,431	KG	2 Classes	Google/Wiki/WSDM	En
UFC (Kim and Choi, 2020)	Triple	1,759	KG	2 Classes	Wiki	En
LieDetect (Mihalcea and Strapparava, 2009)	Passage	600	✗	2 Classes	News	En
FakeNewsAMT (Pérez-Rosas et al., 2018)	Passage	680	✗	2 Classes	News	En
FEVER (Thorne et al., 2018a)	Statement	185,445	Text	3 Classes	Wiki	En
HOVER (Jiang et al., 2020)	Statement	26,171	Text	3 Classes	Wiki	En
WikiFactCheck (Sathe et al., 2020)	Statement	124,821	Text	2 Classes	Wiki	En
VitaminC (Schuster et al., 2021)	Statement	488,904	Text	3 Classes	Wiki	En
TabFact (Chen et al., 2020)	Statement	92,283	Table	2 Classes	Wiki	En
InfoTabs (Gupta et al., 2020)	Statement	23,738	Table	3 Classes	Wiki	En
Sem-Tab-Fact (Wang et al., 2021)	Statement	5,715	Table	3 Classes	Wiki	En
FEVEROUS (Aly et al., 2021)	Statement	87,026	Text/Table	3 Classes	Wiki	En
ANT (Khouja, 2020)	Statement	4,547	✗	3 Classes	News	Ar
DanFEVER (Nørregaard and Derczynski, 2021)	Statement	6,407	Text	3 Classes	Wiki	Da

From [6]

Current state of affairs

- Lots of room for improvement
  - On AVERITEC (one of the newest datasets [16])

Rank	Team	Question Score	Evidence Score	Verdict Accuracy	Justification Score
1	HUMANE	0.8897	0.5358	0.5455	0.5557
2	ADA-AGGR	0.3701	0.4629	0.5369	0.4331
3	AIC CTU	0.8065	0.3251	0.3466	0.3043
4	XxP	0.3902	0.2703	0.2557	0.198
5	teamName	0.6618	0.2285	0.2557	0.2162
6	REVEAL	0.6317	0.2771	0.2358	0.1348
7	fv	0.2885	0.1626	0.1591	0.1306
8	Baseline	0.5545	0.1707	0.1136	0.1322

Resources

- Many surveys, papers, etc.
  - Ex: *"A survey of automated fact-checking"* [6]
- Many shared tasks & a dedicated workshop
  - FEVER Workshop: `https://fever.ai/workshop.html`
  - AVerImaTeC Shared Task: `https://fever.ai/task.html`
- Github (though not complete):
  `https://github.com/Cartus/Automated-Fact-Checking-Resources`

Questions?

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